Instance-Enriched Semantic Maps for Visual Language Navigation

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ABSTRACT

Visual Language Navigation (VLN) aims to enable an embodied agent to navigate within complex environments by following natural language instructions. Achieving this goal requires agents to jointly perceive their surroundings and interpret human language. Recent approaches build semantic spatial maps and leverage Large Language Models (LLMs) for reasoning and decision making. Despite recent advances, existing VLN systems still suffer from a lack of instance-level object detail and robustness to diverse user queries, which limits reliable navigation in complex indoor environments. To address these fundamental limitations, we propose Instance-Enriched Semantic Maps, a unified framework with three key contributions: (1) A novel instance-level 2.5D mapping approach that constructs maps from RGB observations via open-vocabulary panoptic segmentation, crucially preserving vertical distinctions to capture overlapping instances along the height. (2) An integrated room-level semantic segmentation method that combines the Line Segment Detector (LSD) with vision-language inference to provide spatial context. (3) A comprehensive instance captioning system with Large Language Models (LLMs) for rich natural language descriptions that retain detailed attributes beyond simple categorical labels. Our framework demonstrates significant improvements across multiple dimensions. It consistently outperforms baselines in semantic mapping and room segmentation, while reducing memory usage by approximately 95% compared to 3D scene-graph approaches. In navigation experiments, our method achieves over 6% average improvement across diverse query types in both object retrieval and navigation success.

1. Introduction

Achieving robust operation in complex and dynamic environments remains a fundamental challenge for autonomous robots (Kawaharazuka, Matsushima, Gambardella, Guo, Paxton and Zeng, 2024; Radosavovic, Xiao, Zhang, Darrell, Malik and Sreenath, 2024; Yang, Kang, Lee and Kim, 2025; Zhang, Hu, Xiao and Zhang, 2022). A key step toward overcoming this challenge lies in enabling robots to understand and act upon human language, which allows them to interpret high-level intent and adapt their behavior accordingly (Arumugam, Karamcheti, Gopalan, Wong and Tellex, 2017; Huang, Mees, Zeng and Burgard, 2023; Wang, Liu, He, Li, Yan, Chen and Chen, 2024). In this context, Visual Language Navigation (VLN) has emerged as a promising approach that enables embodied agents to follow natural language instructions in photo-realistic environments (Anderson, Wu, Teney, Bruce, Johnson, Sünderhauf, Reid, Gould and van den Hengel, 2018; Savva, Kadian, Maksymets, Zhao, Wijmans, Jain, Straub, Liu, Koltun, Malik, Parikh and Batra, 2019; Kolve, Mottaghi, Han, VanderBilt, Weihs, Herrasti, Deitke, Ehsani, Gordon, Zhu et al., 2017). Recent approaches (Huang et al., 2023; Jatavallabhula, Kuwajerwala, Gu, Omama, Iyer, Saryazdi, Chen, Maalouf, Li, Keetha, Tewari, Tenenbaum, de Melo, Krishna, Paull, Shkurti and Torralba, 2023; Werby, Huang, Büchner, Valada and Burgard, 2024) leverage pre-trained Vision-Language Models (VLMs) to construct semantic maps and achieve zero-shot navigation without task-specific fine-tuning, demonstrating the growing potential of open-vocabulary understanding in embodied agents.

Despite recent progress, existing methods remain inherently instance-agnostic, often failing to distinguish between multiple instances of the same object category. This limitation severely restricts their ability to interpret fine-grained or attribute-based queries, which are essential for robust real-world navigation. While recent efforts have explored instance-level semantic mapping (Nanwani, Agarwal, Jain, Prabhakar, Monis, Mathur, Jatavallabhula, Abdul Hafez, Gandhi and Krishna, 2023; Huang, Zhang, Zhao, Wu and Liu, 2025), their representations remain limited in spatial structure and computational efficiency, lacking the expressiveness required for open-vocabulary reasoning.

To address these limitations, we propose Instance-Enriched Semantic Maps, a unified instance-level semantic mapping framework that incorporates detailed object-level information to support fine-grained and open-vocabulary navigation. As illustrated in Fig. 1, this approach comprises three components for mapping and one component for navigation.

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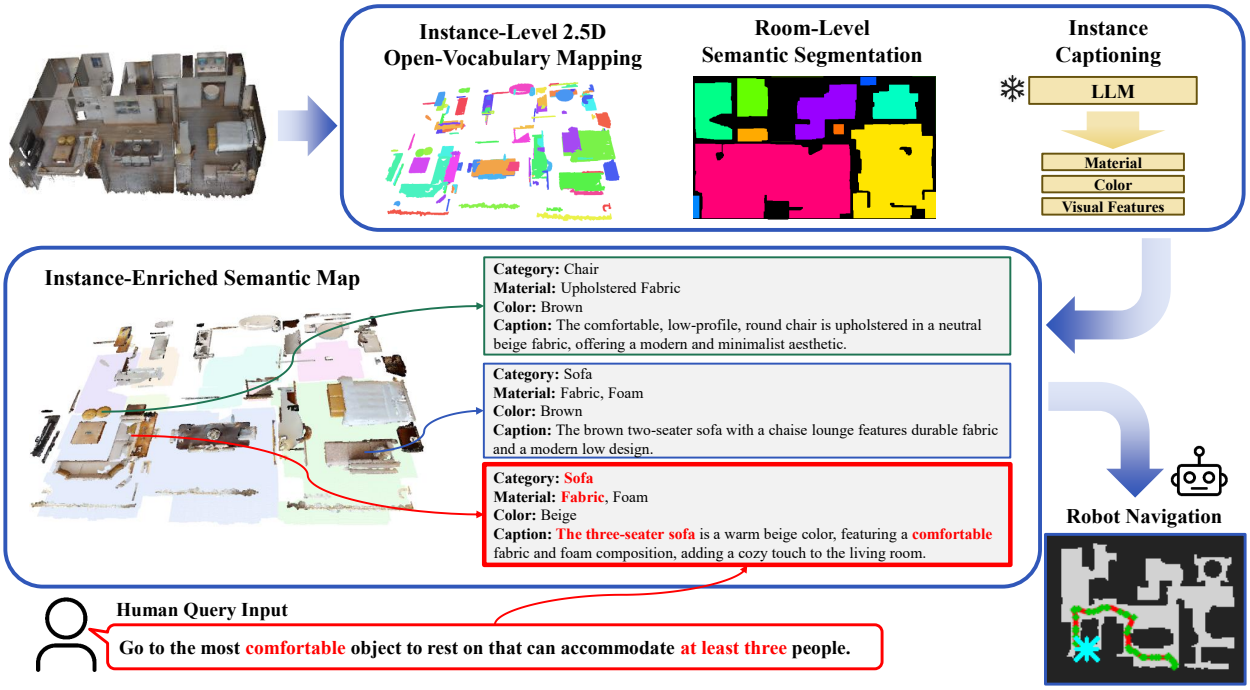


Figure 1: Instance-Enriched Semantic Map is an open-vocabulary 2.5D map representation built upon Instance-Level 2.5D Open-Vocabulary Mapping, Room-Level Semantic Segmentation, and Instance Captioning, thereby enabling robust zero-shot navigation from diverse natural language instructions.

First, an instance-level 2.5D map is built by applying SEEM (Zou, Yang, Zhang, Li, Li, Wang, Wang, Gao and Lee, 2023) for open-vocabulary panoptic segmentation on RGB observations and integrating instances across frames in a global coordinate frame. The 2.5D representation preserves vertical structure and allows overlapping instances along height, which reduces semantic ambiguity common in 2D projections. Second, room-level semantic segmentation is performed by combining the Line Segment Detector (LSD) (Grompone von Gioi, Jakubowicz, Morel and Randall, 2012) with CLIP-based (Radford, Kim, Hallacy, Ramesh, Goh, Agarwal, Sastry, Askell, Mishkin, Clark, Krueger and Sutskever, 2021) vision-language inference to assign instances to spatial regions. Third, LLaMA 3.2 Vision (Dubey, Jauhri, Pandey, Kadian, Al-Dahle, Letman, Mathur, Schelten, Yang, Fan, Goyal, Hartshorn, Yang, Mitra, Sravankumar, Korenev, Hinsvark, Rao, Zhang, Rodriguez, Gregerson, Spataru, Rozière, Biron, Tang, Chern, Caucheteux, Nayak, Bi, Marra, McConnell, Keller, Touret, Wu, Wong, Ferrer, Nikolaidis, Allonsius, Song, Pintz, Livshits, Esiobu, Choudhary, Mahajan, Garcia-Olano, Perino, Hupkes, Lakomkin, AlBadawy, Lobanova, Dinan, Smith, Radenovic, Zhang, Synnaeve, Lee, Anderson, Nail, Mialon, Pang, Cucurell, Nguyen, Korevaar, Xu, Touvron, Zarov, Ibarra, Kloumann, Misra, Evtimov, Copet, Lee, Geffert, Vranes, Park, Mahadeokar, Shah, van der Linde, Billock, Hong, Lee, Fu, Chi, Huang, Liu, Wang, Yu, Bitton, Spisak, Park, Rocca, Johnstun, Saxe, Jia, Alwala, Upasani, Plawiak, Li, Heafield, Stone et al., 2024) generates descriptive captions per instance, capturing properties such as color, material, and texture beyond discrete labels. Finally, during navigation, a Large Language Model (LLM) retrieves target instances that satisfy user queries to select candidate destinations, and the planner drives the robot to the selected goal.

The main contributions of this paper are as follows:

- **Instance-level 2.5D rich information mapping**: A comprehensive mapping framework is introduced that stores diverse semantic attributes at the instance level, such as spatial context, visual properties, and descriptive captions, thereby significantly improving navigation accuracy through enhanced object understanding and disambiguation.
- **Robust query processing via LLM-based target selection**: LLM-based target selection methodology delivers robust handling of diverse query formulations by directly providing mapped instance information to language

models for candidate inference. This design supersedes conventional keyword extraction methods, leveraging semantic reasoning to maintain consistent performance across varied query expressions and formulations.

- **Memory-efficient semantic representation**: A 2.5D representation that combines instance-centric mapping with LLM-based captioning achieves approximately 95% storage reduction compared to 3D scene-graph approaches. Despite the compactness, it preserves sufficient spatial information for precise navigation, as validated through experiments on semantic mapping accuracy and navigation performance.

The remainder of this paper is organized as follows. Sec.2 reviews related work. Sec. 3 details the instance-enriched mapping framework, including construction of the 2.5D top-down map from segmented instances, room-level segmentation, and descriptive captioning, and then explains goal-directed navigation with obstacle-aware instance selection guided by language input. Sec. 4 presents experimental results in simulated environments, evaluating mapping accuracy, object retrieval, and navigation performance under complex language queries. Sec. 5 concludes with a summary and future directions.

2. Related Work

Classical navigation methods rely on Simultaneous Localization and Mapping (SLAM) techniques to build geometric maps for motion planning. SLAM++ (Salas-Moreno, Newcombe, Strasdat, Kelly and Davison, 2013) extends classical SLAM with 3D object models for object-level reasoning. Early semantic mapping work integrated 2D CNN-based predictions into SLAM frameworks (McCormac, Handa, Davison and Leutenegger, 2017), with subsequent methods adding real-time instance segmentation (Runz, Buffier and Agapito, 2018; Mccormac, Clark, Bloesch, Davison and Leutenegger, 2018; Xu, Li, Tzoumanikas, Bloesch, Davison and Leutenegger, 2019). However, these approaches are limited by pre-defined semantic classes and fail to capture fine-grained spatial relationships. Learning-based approaches optimize end-to-end policies mapping language commands to actions, but require large-scale training datasets and lack explicit spatial reasoning.

VLMs pre-trained on large-scale Internet data enable robots to ground natural language instructions in visual observations without fine-tuning (Sumers, Marino, Ahuja, Fergus and Dasgupta, 2023). These models support open-vocabulary object recognition and have been integrated into exploration-based navigation frameworks (Gadre, Wortsman, Ilharco, Schmidt and Song, 2023; Shah, Osiński, Ichter and Levine, 2022). Such approaches primarily use VLMs to evaluate alignment between visual inputs and goal descriptions, but remain disconnected from spatial mapping. Recent works propose zero-shot techniques for semantic spatial maps by integrating visual landmarks with language (Huang et al., 2023; Chen, Xia, Ichter, Rao, Gopalakrishnan, Ryoo, Stone and Kappler, 2023). Large Language Models (LLMs) decompose high-level instructions into actionable subgoals (Yuan, Wang and Waslander, 2025; Zhao, Lu, Kim, Fu, Zhang, Wu, Li, Ma, Han, Finn, Handa, Lin, Wetzstein, Liu and Xiang, 2025; Ichter, Brohan, Chebotar, Finn, Hausman, Herzog, Ho, Ibarz, Irpan, Jang, Julian, Kalashnikov, Levine, Lu, Parada, Rao, Sermanet, Toshev, Vanhoucke, Xia, Xiao, Xu, Yan, Brown, Ahn, Cortes, Sievers, Tan, Xu, Reyes, Rettinghouse, Quiambao, Pastor, Luu, Lee, Kuang, Jesmonth, Jeffrey, Ruano, Hsu, Gopalakrishnan, David, Zeng and Fu, 2022; Driess, Xia, Sajjadi, Lynch, Chowdhery, Ichter, Wahid, Tompson, Vuong, Yu, Huang, Chebotar, Sermanet, Duckworth, Levine, Vanhoucke, Hausman, Toussaint, Greff, Zeng, Mordatch and Florence, 2023). However, these methods lack the spatial reasoning capabilities needed for complex navigation scenarios with diverse linguistic expressions.

Recent zero-shot mapping approaches utilize different Vision-Language Model (VLM) architectures with varying trade-offs. VLMaps employs LSeg (Li, Weinberger, Belongie, Koltun and Ranftl, 2022) for semantic mapping, but suffers from noisy per-pixel classification within the same object instances. The widely used SAM+CLIP combination (Kirillov, Mintun, Ravi, Mao, Rolland, Gustafson, Xiao, Whitehead, Berg, Lo, Dollar and Girshick, 2023; Radford et al., 2021) benefits from strong segmentation and image-language alignment. However, it struggles with object-level awareness, as SAM generates segments that include irrelevant or semantically meaningless regions, while CLIP extracts semantic features from the entire image, which can cause background noise and reduce semantic precision. Some approaches utilize panoptic segmentation models such as SEEM (Zou et al., 2023) to provide region-level features, with Open-Fusion (Yamazaki, Hanyu, Vo, Pham, Tran, Doretto, Nguyen and Le, 2024) extending this idea toward real-time 3D representations. To improve open-vocabulary scene understanding, ConceptFusion (Jatavallabhula et al., 2023) incorporates multimodal features into 3D reconstruction, while hierarchical scene-graph approaches (Gu, Kuwajerwala, Morin, Jatavallabhula, Sen, Agarwal, Rivera, Paul, Ellis, Chellappa, Gan, de Melo, Tenenbaum, Torralba, Shkurti and Paull, 2024; Werby et al., 2024) further support flexible queries and relational reasoning. Despite

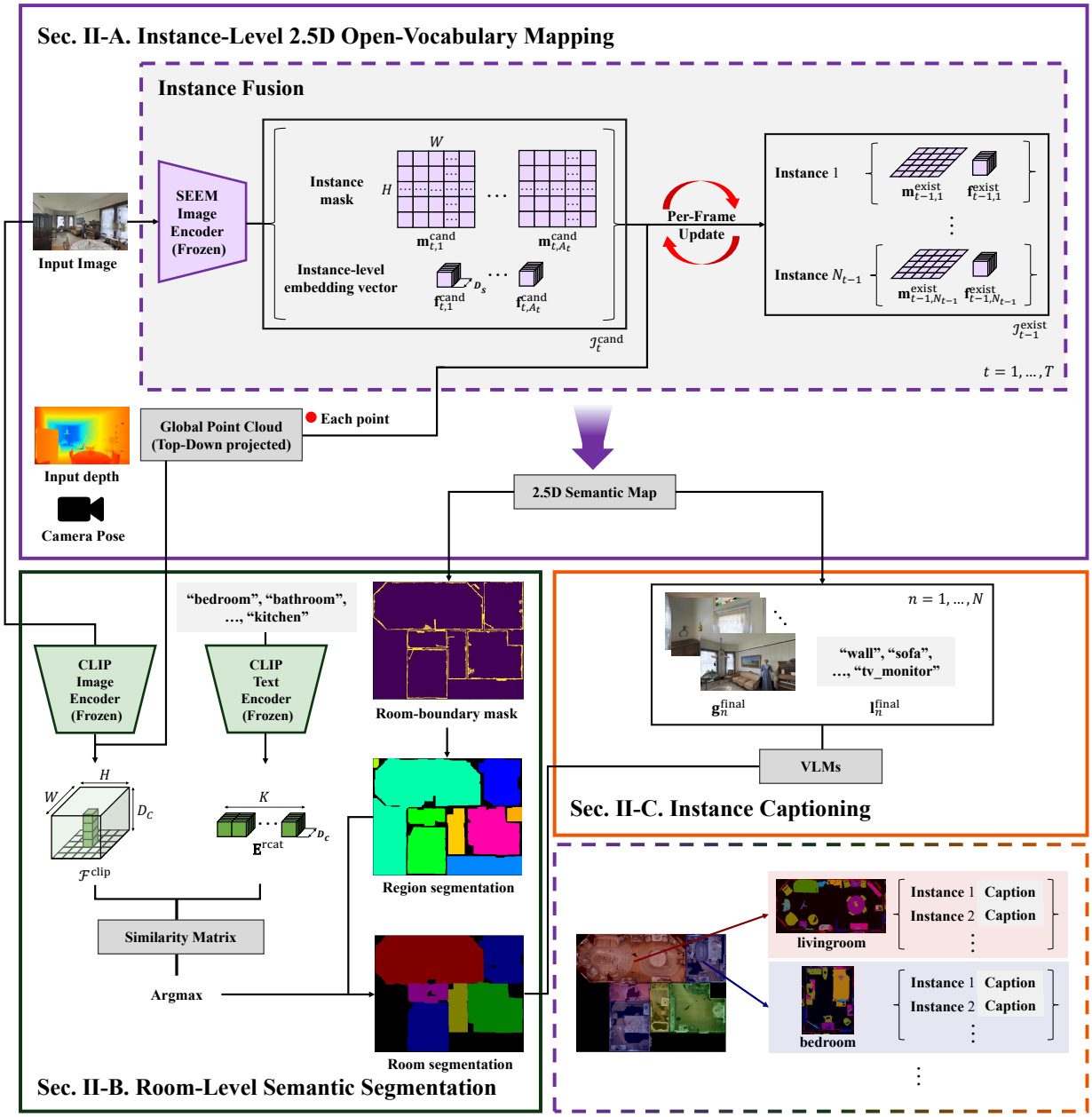


Figure 2: System overview. The proposed framework consists of three main components: (i) Instance-Level 2.5D Open-Vocabulary Mapping (Sec. 3.1), where per-frame instance masks and embeddings obtained from SEEM are projected into a global 2.5D semantic map through instance fusion; (ii) Room-Level Semantic Segmentation (Sec. 3.2), where CLIP-based vision-language grounding is combined with structural room boundary masks to produce region-consistent room categorization; and (iii) Instance Captioning (Sec. 3.3), where representative frames are selected for each instance and passed through vision-language models to generate detailed textual descriptions enriched with room context. These components yield a unified 2.5D semantic map augmented with open-vocabulary labels and natural language descriptions, supporting open-ended navigation and interaction tasks.

their strengths in providing rich spatial representations, these 3D-based methods require considerable storage, which limits scalability in large environments. Alternative 2D-based instance-aware VLN methods (Nanwani et al., 2023;

Huang et al., 2025) exist, but they fail to handle vertical overlap, leading to blended representations and inaccurate features. These limitations highlight the need for frameworks that integrate instance-level semantics with natural language descriptions while maintaining computational efficiency.

3. Methodology

A multi-stage pipeline generates a semantic, instance-centric map from RGB-D inputs to support language-based navigation. The process begins by deriving instance segmentation masks from RGB inputs using a VLM, which are subsequently merged across frames to create a temporally consistent instance map. Based on the merged instances, architectural structures are identified to generate a room separation mask, which is then combined with an obstacle map to define navigable regions. Vision-language embeddings are employed to assign room labels, resulting in a semantic room map. Finally, each object instance is described with captions generated from visual features aggregated across multiple frames. The resulting multi-layer map supports flexible and abstract language-based navigation queries. An overview of the full pipeline is shown in Fig. 2.

3.1. Instance-level 2.5D open-vocabulary mapping

Instance-level open-vocabulary information is mapped into a global 2.5D representation through a five-step pipeline. Candidate masks and embeddings are extracted from RGB-D frames, associated with prior instances by geometric and semantic similarity, and fused into consistent entries. Our instance association and fusion process assumes accurate and globally consistent camera poses, obtained either from the simulator or from a reliable SLAM estimation. The final output consists of (i) a 2.5D map $\mathcal{M}$, constructed by integrating object-wise occupancy regions, where each grid cell stores indices of overlapping instances, thus preserving multi-occupancy per cell and (ii) an instance dictionary $\mathcal{I}$ that consolidates semantic features and category labels.

3.1.1. Instance-Centric Representation

From a sequence of RGB-D observations, the panoptic segmentation module of SEEM extracts candidate instance masks together with their open-vocabulary feature embeddings at each time step. Each candidate instance is reconstructed in 3D from depth data, transformed into the global coordinate frame using the camera pose, and projected top-down onto a grid map.

At time step t, the candidate instances are organized into the following dictionary:

$$\mathcal{I}_t^{\text{cand}} = \left\{ a \mapsto \mathbf{I}_{t,a}^{\text{cand}} \right\}_{a=1,\ldots,A_t}, \tag{1}$$

where each entry is defined as

$$\mathbf{I}_{t,a}^{\text{cand}} = \left(\mathbf{m}_{t,a}^{\text{cand}}, \mathbf{f}_{t,a}^{\text{cand}}\right). \tag{2}$$

Here, $\mathbf{m}_{t,a}^{\text{cand}} \in \{0,1\}^{H\times W}$ denotes the top-down binary mask of candidate a, and $\mathbf{f}_{t,a}^{\text{cand}} \in \mathbb{R}^{D_S}$ represents its feature embedding. D_S denotes the dimensionality of SEEM feature embeddings, A_t denotes the number of candidate instances obtained from the t-th image frame, and H, W denote the height and width of the top-down grid, respectively.

3.1.2. Instance Association

Newly detected candidate instances are compared against the N_{t-1} instances accumulated up to time step $t-1$. The information obtained previously is organized into the following dictionary:

$$\mathcal{I}_{t-1}^{\text{exist}} = \left\{ n \mapsto \mathbf{I}_{t-1,n}^{\text{exist}} \right\}_{n=1,\ldots,N_{t-1}}, \tag{3}$$

where each entry is defined as

$$\mathbf{I}_{t-1,n}^{\text{exist}} = \left(\mathbf{m}_{t-1,n}^{\text{exist}}, \mathbf{f}_{t-1,n}^{\text{exist}}, \mathbf{g}_{t-1,n}^{\text{exist}}, \bar{h}_{t-1,n}^{\text{exist}}, w_{t-1,n}^{\text{exist}}\right). \tag{4}$$

Here, $\mathbf{m}_{t-1,n}^{\text{exist}} \in \{0,1\}^{H\times W}$ denotes the 2D top-down mask of instance n, and $\mathbf{f}_{t-1,n}^{\text{exist}} \in \mathbb{R}^{D_S}$ is its aggregated feature embedding. $\mathbf{g}_{t-1,n}^{\text{exist}} = [g_{t-1,n,1}^{\text{exist}}, \ldots, g_{t-1,n,V}^{\text{exist}}]$ is the frame-index list of instance n, where V is the total number of frames

in which the instance appears, and $g^{\text{exist}}_{t-1,n,v}$ denotes a specific frame index. $\bar{h}^{\text{exist}}_{t-1,n}$ represents the mean height of instance n, computed over its occupied grid cells, and $w^{\text{exist}}_{t-1,n} \in \mathbb{N}$ denotes the weight corresponding to the number of frames in which the instance has been observed up to time step $t-1$.

Matching a newly observed candidate instance $\mathbf{I}^{\text{cand}}_{t,a}$ to an existing instance $\mathbf{I}^{\text{exist}}_{t-1,n}$ relies on geometric and semantic similarities. The geometric similarity ϕ_{geo} is defined as

$$\phi_{\text{geo}}\big(\mathbf{m}_\alpha, \mathbf{m}_\beta\big) = \max\Big(\text{o}(\mathbf{m}_\alpha, \mathbf{m}_\beta), \text{o}(\mathbf{m}_\beta, \mathbf{m}_\alpha)\Big), \tag{5}$$

where $\text{o}(\mathbf{m}_\alpha, \mathbf{m}_\beta)$ is the overlap ratio, which is the intersection area normalized by the area of $\mathbf{m}_\alpha$. Compared with Intersection Over Union (IoU), max-overlap mitigates the effects of single-view truncation and top-down compression, allowing matches to be correctly identified even when the overlap is below a predefined threshold. The semantic similarity ϕ_{sem} is given by

$$\phi_{\text{sem}}\big(\mathbf{f}_\alpha, \mathbf{f}_\beta\big) = \mathbf{f}_\alpha \mathbf{f}_\beta^\top, \tag{6}$$

which equals the cosine similarity. The overall similarity is defined as a convex combination:

$$\phi = \omega_{\text{geo}}\phi_{\text{geo}} + \omega_{\text{sem}}\phi_{\text{sem}}, \qquad \omega_{\text{geo}} + \omega_{\text{sem}} = 1. \tag{7}$$

For each candidate entry $\mathbf{I}^{\text{cand}}_{t,a}$, the matched instance index i is selected from the N_{t-1} accumulated entries in $\mathcal{I}^{\text{exist}}_{t-1}$ that maximizes ϕ. The selected instance satisfy both thresholds $\phi_{\text{geo}} > \delta_{\text{geo}}$ and $\phi_{\text{sem}} > \delta_{\text{sem}}$:

$$i = \underset{n=1,\dots,N_{t-1}}{\text{argmax}} \ \phi\big(\mathbf{I}^{\text{cand}}_{t,a}, \mathbf{I}^{\text{exist}}_{t-1,n}\big). \tag{8}$$

3.1.3. Instance Fusion

Let $\Omega = \{1, \dots, H\} \times \{1, \dots, W\}$ denote the discrete grid domain of the top-down map. If a candidate instance $\mathbf{I}^{\text{cand}}_{t,a}$ does not match any existing entry, it is added as a new detection to the dictionary $\mathcal{I}^{\text{exist}}_t$. Otherwise, matched entries are fused as follows. The mask is updated by applying a logical OR between the candidate mask and the existing mask:

$$\mathbf{m}^{\text{exist}}_{t,i}(x, y) = \mathbf{m}^{\text{cand}}_{t,a}(x, y) \vee \mathbf{m}^{\text{exist}}_{t-1,i}(x, y), \quad (x, y) \in \Omega. \tag{9}$$

The feature embedding is updated via a weighted average of the previous embedding and the candidate embedding, where the previous weight is used as the factor. Once a match is confirmed, the weight is incremented by one:

$$\mathbf{f}^{\text{exist}}_{t,i} = \frac{w^{\text{exist}}_{t-1,i}\, \mathbf{f}^{\text{exist}}_{t-1,i} + \mathbf{f}^{\text{cand}}_{t,a}}{w^{\text{exist}}_{t-1,i} + 1}, \quad w^{\text{exist}}_{t,i} = w^{\text{exist}}_{t-1,i} + 1. \tag{10}$$

The frame-index list is extended by appending the current frame index, and its items are sorted by occupied pixel area. The mean height is updated using the same weighted averaging scheme as the feature embedding.

After all frames are processed, the dictionary $\mathcal{I}^{\text{exist}}_T$ is obtained, where T denotes the total number of time steps. A second association–fusion pass, using the same procedure as above but applied within $\mathcal{I}^{\text{exist}}_T$, merges long-range duplicates arising from occlusion, pose drift, or view changes and enforces global instance consistency. This yields the definitive dictionary $\mathcal{I}^{\text{final}} = \{\, n \mapsto \mathbf{I}^{\text{final}}_n \,\}_{n=1,\dots,N}$, where N denotes the number of final entries.

3.1.4. Instance-Level 2.5D Map Creation

Per-instance masks are aggregated to construct the global 2.5D map:

$$\mathcal{M} = \{\, \mathcal{M}_{x,y} \mid (x, y) \in \Omega \,\}, \quad \mathcal{M}_{x,y} = [n_1, \dots, n_K], \tag{11}$$

where each cell $\mathcal{M}_{x,y}$ stores the indices of all instances occupying that cell, and K denotes the number of occupying instances at $\mathcal{M}_{x,y}$.

This multi-occupancy per cell motivates the 2.5D designation, as multiple instances can be stacked within a single grid cell. After the map is constructed, the per-instance mask $\mathbf{m}^{\text{final}}_n$ contained in each $\mathbf{I}^{\text{final}}_n$ is discarded to avoid redundancy.

3.1.5. Open-Vocabulary Instance Categorization

Finally, instance feature embeddings are categorized using an open-vocabulary approach. Given a language-category list $\mathcal{L}^{\text{icat}} = [\mathbf{l}_1^{\text{icat}}, \mathbf{l}_2^{\text{icat}}, \dots, \mathbf{l}_J^{\text{icat}}]$, where $\mathbf{l}_j^{\text{icat}}$ denotes the j-th category text and J is the number of categories, textual embeddings $[\mathbf{e}_1^{\text{icat}}, \mathbf{e}_2^{\text{icat}}, \dots, \mathbf{e}_J^{\text{icat}}]$, $\mathbf{e}_j^{\text{icat}} \in \mathbb{R}^{D_S}$ are generated with a pre-trained SEEM text encoder. These embeddings constitute the matrix $\mathbf{E}^{\text{icat}} \in \mathbb{R}^{J \times D_S}$, whose rows correspond to category embeddings. In parallel, embeddings $[\mathbf{f}_1^{\text{final}}, \dots, \mathbf{f}_N^{\text{final}}]$ from $\mathcal{I}^{\text{final}}$ form the matrix $\mathbf{F}^{\text{final}} \in \mathbb{R}^{N \times D_S}$, where each row represents the embedding of an instance.

An instance-to-category similarity matrix is then computed as $\mathbf{Z} = \mathbf{F}^{\text{final}}(\mathbf{E}^{\text{icat}})^\top$, where $\mathbf{Z} \in \mathbb{R}^{N \times J}$ stores similarity scores for all pairs. The category index for instance n is obtained by

$$c_n = \underset{j=1,\dots,J}{\operatorname{argmax}} \mathbf{Z}_{nj}, \tag{12}$$

where c_n indicates the position of the selected category in $\mathcal{L}^{\text{icat}}$. Using this index, the corresponding category text is retrieved and stored as the label $\mathbf{l}_n^{\text{final}}$ within the definitive entry $\mathbf{I}_n^{\text{final}}$.

Through this pipeline, the instance-level 2.5D open-vocabulary mapping is completed, yielding two final outputs: the global 2.5D map $\mathcal{M} = \{\mathcal{M}_{x,y} \mid (x, y) \in \Omega\}$ and the definitive instance dictionary $\mathcal{I}^{\text{final}} = \{n \mapsto \mathbf{I}_n^{\text{final}}\}_{n=1,\dots,N}$, with each entry defined as $\mathbf{I}_n^{\text{final}} = (\mathbf{f}_n^{\text{final}}, \mathbf{g}_n^{\text{final}}, \bar{h}_n^{\text{final}}, w_n^{\text{final}}, \mathbf{l}_n^{\text{final}})$.

3.2. Room-level semantic segmentation

This section presents a two-step approach to generate room segmentation maps. First, individual room instances are separated using structural object masks. Then, a semantic mask is generated by computing the similarity between CLIP visual and linguistic features. This mask is combined with the structural map for region-wise aggregation.

3.2.1. Instance-Level Room Separation

The instance segmentation masks produced by SEEM serve as the basis for extracting structural elements that typically delineate room boundaries such as walls, doors, and large fixed furniture. To ensure that segmentation occurs only within navigable areas, an obstacle map is incorporated to exclude non-traversable regions. This obstacle map is a binary representation of the environment that marks non-traversable regions, based on depth sensing and occupancy estimation, and its construction is explicitly described in Sec. 3.4. The selected masks are merged into a room-boundary mask $\mathbf{m}^{\text{room}}$, a binary representation in which boundary pixels are set to 1 and all other pixels are set to 0. This binary format enhances structural contrast and serves as an input for subsequent line detection.

The LSD (Grompone von Gioi et al., 2012) is applied to extract wall-like linear structures. For each pixel (x, y) of the boundary mask $\mathbf{m}^{\text{room}}$, the horizontal and vertical gradients are computed as

$$z_x(x, y) = \mathbf{m}^{\text{room}}(x + 1, y) - \mathbf{m}^{\text{room}}(x - 1, y), \tag{13}$$

$$z_y(x, y) = \mathbf{m}^{\text{room}}(x, y + 1) - \mathbf{m}^{\text{room}}(x, y - 1). \tag{14}$$

The level-line angle θ_{xy}, perpendicular to the gradient orientation, is then calculated as

$$\theta_{xy} = \arctan\left(\frac{z_y}{z_x}\right). \tag{15}$$

Then, pixels whose level-line angles differ by less than a threshold τ from a candidate line direction θ_0 are grouped into connected regions, referred to as Line Support Regions (LSRs). A pixel (x, y) belongs to the $\mathcal{S}^{\text{cand}} = \{\mathbf{s}_c^{\text{cand}}\}_{c=1\dots C}$, where C denotes the number of LSR if $\left|\theta_{xy} - \theta_0\right| < \tau$. Each region $\mathbf{r}_c$ in LSR is approximated by a minimum bounding rectangle, representing a candidate line segment. For each candidate rectangle, the alignment ratio r is computed as

$$r = \frac{o}{q}, \tag{16}$$

where o is the number of pixels aligned with the rectangle's orientation and q is the total number of pixels in the rectangle.

To reject false detections, the Number of False Alarms (NFA) metric is computed as

$$\mathrm{NFA}(\mathbf{r}_c) = N_{\text{tests}} \cdot \sum_{j=o}^{q} \binom{q}{j} p^j (1-p)^{q-j}, \tag{17}$$

where N_{tests} is the number of tests conducted and p is the probability, modeled as a binomial distribution. Only line segments satisfying NFA ≤ 1 are retained.

Detected line segments shorter than a length threshold are discarded, and adjacent collinear segments are merged to improve continuity. The accepted line segments are rasterized into a boundary mask $\mathbf{m}^{\text{room}}$, which is refined via morphological dilation and closing operations to fill gaps and connect fragmented segments. Finally, connected component analysis is performed on the refined mask to extract individual room instances. Each component is assigned a unique instance ID, resulting in a structural room instance map $\mathcal{S}^{\text{final}} = \{\mathbf{s}_b^{\text{final}}\}_{b=1\ldots B}$, where B denotes the number of room mask. While this map delineates individual rooms, it lacks semantic information about room types, thereby requiring a second stage for semantic labeling.

3.2.2. Semantic Labeling

Dense image-text embedding maps are extracted using the CLIP VLM to assign semantic labels to each room instance. Let $\mathcal{L}^{\text{rcat}} = [\mathbf{l}_1^{\text{rcat}}, \mathbf{l}_2^{\text{rcat}}, \ldots, \mathbf{l}_K^{\text{rcat}}]$ denote the set of predefined room-type categories (e.g., kitchen, bedroom, living room), where K is the number of room types. Using the same RGB observations as in the previous stage, the CLIP image encoder produces a dense embedding map $\mathcal{F}^{\text{clip}} \in \mathbb{R}^{H\times W\times D_C}$, where each pixel (u, v) is represented by a D_C-dimensional feature vector aligned with natural language concepts. Similarly, the CLIP text encoder converts each room category $\mathbf{l}_k^{\text{rcat}}$ into a text embedding vector $\mathbf{e}_k^{\text{rcat}} \in \mathbb{R}^{D_C}$, forming the embedding matrix $\mathbf{E}^{\text{rcat}} \in \mathbb{R}^{K\times D_C}$. The embedding map $\mathcal{F}^{\text{clip}}$ is flattened into a matrix $\mathbf{Q}^{\text{clip}} \in \mathbb{R}^{HW\times D_C}$. The similarity matrix between each pixel and text embedding is then computed as $\mathbf{S}^{\text{clip}} = \mathbf{Q}^{\text{clip}}(\mathbf{E}^{\text{rcat}})^\top$ where $\mathbf{S}^{\text{clip}} \in \mathbb{R}^{HW\times K}$. The most relevant room label for each pixel n is obtained via

$$\mathbf{r} = \underset{k=1\ldots K}{\operatorname{argmax}}\, \mathbf{S}_{nk}^{\text{clip}}. \tag{18}$$

The resulting vector $\mathbf{r} \in \mathbb{R}^{HW}$ is reshaped to form an initial room semantic segmentation map $\mathbf{R} \in \mathbb{R}^{H\times W}$. However, direct pixel-wise classification is sensitive to the agent's viewpoint, often resulting in noisy predictions that do not consistently reflect the actual room context. To mitigate this issue, the previously generated structural room instance map $\mathcal{S}^{\text{final}}$ is used to guide the semantic labeling process. For each room segment $\mathbf{s}_b^{\text{final}}$, the final label for room instance is determined by counting how many times each label appears among all pixels within the segment and selecting the most frequent one:

$$l_b = \underset{k}{\operatorname{argmax}} \left| \{(x, y) \in \mathbf{s}_b^{\text{final}} \mid \mathbf{R}_{xy} = k\} \right|. \tag{19}$$

As a result, the final output of the room-level semantic segmentation stage is defined as

$$\mathcal{R}^{\text{final}} = \{(\mathbf{s}_b^{\text{final}}, l_b)\}_{b=1\ldots B}, \tag{20}$$

where each room segment $\mathbf{s}_b^{\text{final}}$ is assigned the most frequent label l_b aggregated over its pixels. This region-constrained labeling strategy alleviates pixel-level noise and improves the overall reliability of room-type classification.

3.3. Instance captioning

Detailed textual descriptions are generated for each object instance to enrich the semantic map and enable robust responses to open-ended user queries. The captioning pipeline begins after the instance and room segmentation process. For every detected instance n, a set of representative frames $\mathbf{g}_n^{\text{final}}$ is selected during the instance segmentation stage. The selection process takes into account both the confidence of the predicted masks and the frequency of the instance's observations across frames. The selected keyframes are then coupled with the corresponding candidate category $\mathbf{l}_n^{\text{final}}$ obtained from the same stage. Each frame-category pair serves as an input to the LLaMA 3.2 model, which is tasked

```
[
   1: {
        "category": "curtain",
        "room-type": "bedroom",
        "color": "white",
        "material": "lace",
        "captions": "A lace curtain with intricate floral
                     patterns elegantly filters light while
                     providing privacy, featuring vintage charm."
      },
   2: {
        "category": "door",
        "room-type": "living room",
        "color": "dark brown",
        "material": "wood",
        "captions": "The ornate dark wood double door,
                     featuring glass insertions and delicate
                     carvings, exemplifies traditional
                     craftsmanship and adds luxury to the
                     living room."
      },
                                  ...
                                  ...
150: {
        "category": "sink",
        "room-type": "kitchen",
        "color": "white",
        "material": "porcelain",
        "captions": "The single sink made of white porcelain
                     is installed on a countertop, featuring a
                     faucet, adding a functional element to
                     the kitchen."
      },
                                  ...
 ]
```

Figure 3: Example of instance-level caption annotations. Each detected object instance is associated with semantic attributes including category, room type, color, and material.

with producing natural language descriptions. The vision LLM is prompted to identify observable object properties such as color, material, texture, and geometric shape. For spatial grounding, each instance is linked to its associated room category. The spatial location of each object is identified on the 2.5D semantic map, and the instance is labeled with the corresponding room label based on the room-level segmentation map $\mathcal{R}^{\text{final}}$. All instance-level captions are then aggregated through GPT-4 (Achiam, Adler, Agarwal, Ahmad, Akkaya, Aleman, Almeida, Altenschmidt, Altman, Anadkat et al., 2023), which refines them by eliminating redundancy and distilling the essential visual attributes into a compact description. In addition, structured tags are extracted for key attributes.

As illustrated in Fig. 3, the final representation of each instance comprises its room tag providing its spatial context, a natural language caption, and structured key-value tags for efficient indexing and retrieval. This comprehensive representation enables the system to interpret varied open-vocabulary navigation commands, such as "find the small red cup in the kitchen" or "go to the wooden chair in the livingroom". A detailed description of the prompts and captioning procedure can be found in A.

3.4. Open-Vocabulary Navigation from Natural Language

3.4.1. Generating Obstacle Maps

The binary obstacle map is primarily constructed using a height threshold. To account for cases where objects should be treated as obstacles even if their mean height falls outside the specified range (e.g., below $h_{\min}$), an open-vocabulary list of potential obstacle categories $\mathcal{L}^{\text{ocat}} = [\mathbf{l}_1^{\text{ocat}}, \dots, \mathbf{l}_K^{\text{ocat}}]$ is derived from natural-language specifications.

Using the 2.5D map $\mathcal{M}$ and the instance dictionary $\mathcal{I}^{\text{final}}$, each instance with label $\mathbf{I}_n^{\text{final}}$ is treated as an obstacle if either its mean height lies within $[h_{\min}, h_{\max}]$ or its label belongs to $\mathcal{L}^{\text{ocat}}$. The binary map $\mathcal{O} \in \{0,1\}^{H \times W}$ is obtained by marking all grid cells that contain at least one such instance.

3.4.2. Instance-Level Navigation from Natural Language

Leveraging GPT-4 for language understanding, the system identifies the most appropriate target instance to support goal-directed navigation. The model receives a user command and a JSON-formatted description that contains information about all detected instances. The predicted instance ID corresponds to the navigation goal.

4. Experiments

We evaluate the proposed framework from multiple perspectives. First, we measure the accuracy of instance-level 2.5D open-vocabulary mapping and perform an additional analysis to verify the benefits of 2.5D over 2D mapping. Next, we assess the quality of room-level segmentation and examine whether the integrated framework, which combines instance segmentation, room segmentation, and captioning, enhances both goal retrieval from language queries and navigation success rates. Finally, we compare the storage costs of different representations, showing improved efficiency over 3D-based methods.

All experiments are conducted within the Habitat simulator (Savva et al., 2019), using the Matterport3D (MP3D) (Chang, Dai, Funkhouser, Halber, Niebner, Savva, Song, Zeng and Zhang, 2017), Replica (Straub, Whelan, Ma, Chen, Wijmans, Green, Engel, Mur-Artal, Ren, Verma, Clarkson, Yan, Budge, Yan, Pan, Yon, Zou, Leon, Carter, Briales, Gillingham, Mueggler, Pesqueira, Savva, Batra, Strasdat, Nardi, Goesele, Lovegrove and Newcombe, 2019), and Habitat-Semantics dataset (HM3DSem) (Yadav, Ramrakhya, Ramakrishnan, Gervet, Turner, Gokaslan, Maestre, Chang, Batra, Savva, Clegg and Chaplot, 2023) datasets. For fairness, each experiment primarily follows the datasets settings used by the corresponding baselines.

4.1. 2.5D open-vocabulary mapping

4.1.1. Instance-level 2.5D open-vocabulary mapping

The goal is to verify whether the mapping process accurately performs object-level segmentation and mapping without over-segmentation or under-segmentation, and whether open-vocabulary features are appropriately stored at the object level.

Baselines and Dataset: We compare against HOV-SG (Werby et al., 2024) and VLMaps (Huang et al., 2023). For HOV-SG (3D instance-wise features), each object point cloud is projected to a top-down mask for a 2.5D comparison. For VLMaps (2D point-wise features), per-instance descriptors are obtained by masking the feature map with ground-truth instance masks and averaging. Following prior setups, HOV-SG uses CLIP (ViT-H-14), VLMaps uses LSeg (ViT-B-32), and our method uses SEEM (Focal-L). Consistent with HOV-SG, experiments are conducted on the Replica dataset, using five scenes for quantitative evaluation.

Metrics: Following HOV-SG, we adopt the labeling-based AUC^{top} for semantic evaluation at the object level. This metric computes the cumulative ratio of correctly labeled instance matches within the top-k predictions, defined as

$$f(k) = \frac{|S_k|}{|M|}, \quad \text{AUC}^{\text{top}} = \int_0^1 f(x)\, dx, \tag{21}$$

where $|S_k|$ is the number of pairs correctly labeled among the top-k, and $|M|$ is the total number of matched pairs. Here, the horizontal axis is rescaled to $[0, 1]$ by the total number of categories, so that k is mapped to the unit interval, enabling the computation of the Area Under the Curve (AUC). Since $f(k)$ is discrete, we interpolate it over the unit interval to obtain a continuous curve for AUC computation, and the same interpolation procedure is applied consistently to all AUC variants introduced below.

However, instance-level evaluation inherently involves two stages: instance matching and label assignment. The original HOV-SG formulation normalizes by the number of matched pairs $|M|$, which does not explicitly penalize redundant predictions or over-segmentation errors, since the denominator is restricted to the matched subset rather than the entire prediction set.

To overcome this limitation, we introduce additional denominator choices for normalization. Here, normalized refers to the choice of denominator. Prediction-normalized labeling AUC ($\text{AUC}^{\text{top}}_{\text{pred}}$) penalizes redundant or spurious

Table 1
Instance-level semantic labeling performance under different normalization settings on the Replica dataset. Instance Difference, the absolute difference between ground-truth and predicted object counts, is identical for each method regardless of the normalization scheme. Only predicted and ground-truth instances with an IoU greater than 50% are considered as matched pairs.

Normalization	Method (Dim.)	top_1	top_2	top_5	top_{50}	top_{100}	AUC^{top} ↑	Instance Diff ↓
Pred-norm.	HOV-SG (3D)	0.092	0.130	0.146	0.194	0.198	0.186	50.2
	Ours (2.5D)	**0.204**	**0.300**	**0.340**	**0.388**	**0.412**	**0.378**	**5.6**
GT-norm.	HOV-SG (3D)	0.172	0.240	0.268	**0.362**	0.368	0.344	–
	Ours (2.5D)	**0.190**	**0.272**	**0.314**	0.354	**0.374**	**0.346**	–
Matching-norm.	VLMaps + GT mask (2D)	0.070	0.084	0.102	0.462	**1.000**	0.522	–
	HOV-SG (3D)	0.474	0.668	0.748	**0.980**	**1.000**	**0.946**	–
	Ours (2.5D)	**0.502**	**0.722**	**0.832**	0.944	**1.000**	0.924	–

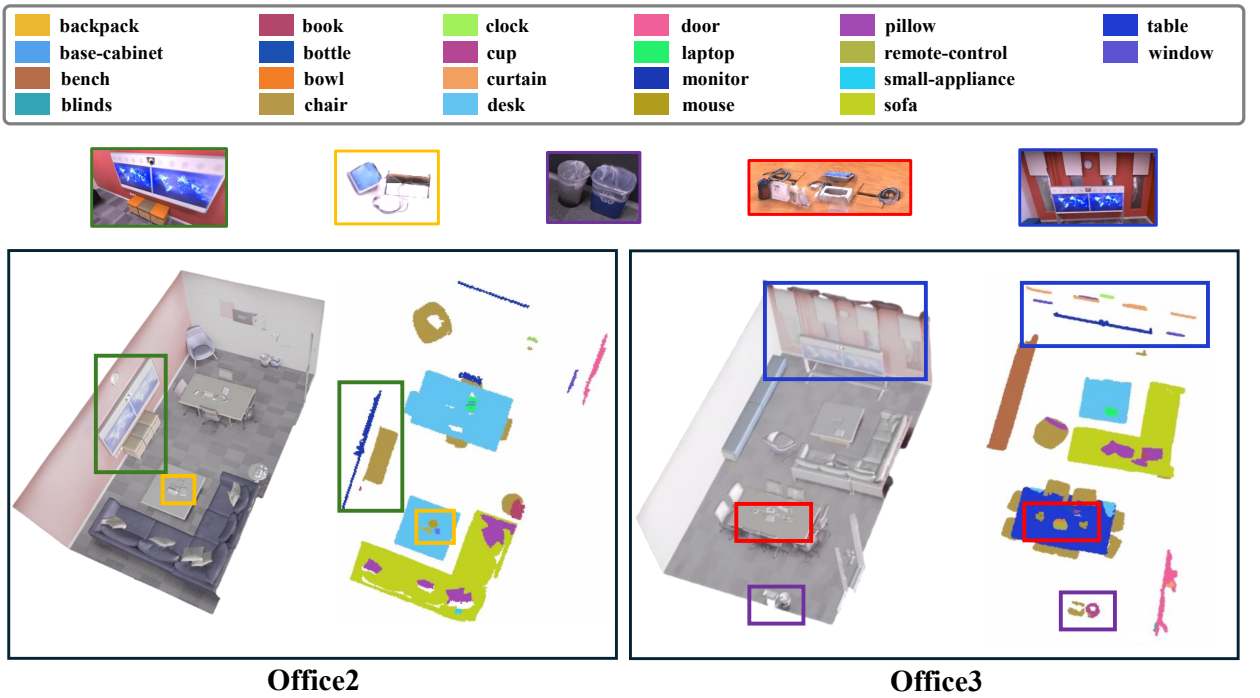


Figure 4: Instance-level 2.5D open-vocabulary mapping results on the Replica dataset. The colored boxes indicate that even small objects are well preserved and accurately mapped without disappearing.

predictions by normalizing with respect to the total number of predicted objects:

$$f_{\text{pred}}(k) = \frac{|S_k|}{|P|}, \quad \text{AUC}^{\text{top}}_{\text{pred}} = \int_0^1 f_{\text{pred}}(x)\,dx, \tag{22}$$

where $|P|$ denotes the total number of predictions. Ground-truth-normalized labeling AUC ($\text{AUC}^{\text{top}}_{\text{gt}}$) penalizes under-segmentation and missed detections by normalizing with respect to the total number of ground-truth objects:

$$f_{\text{gt}}(k) = \frac{|S_k|}{|G|}, \quad \text{AUC}^{\text{top}}_{\text{gt}} = \int_0^1 f_{\text{gt}}(x)\,dx, \tag{23}$$

where $|G|$ denotes the total number of ground-truth objects. Matching-normalized labeling AUC ($\text{AUC}^{\text{top}}_{\text{match}}$) evaluates whether object-level semantics are preserved without additional penalties by normalizing with respect to the number

Table 2
Evaluation of semantic mapping performance against 2D-based models on the MP3D and Replica datasets.

Method	VLM (Backbone)	MP3D				Replica			
		Acc	mAcc	mIoU	F-mIoU	Acc	mAcc	mIoU	F-mIoU
CLIP Map	CLIP (ViT-B-32)	0.106	0.102	0.017	0.026	0.147	0.066	0.012	0.028
VLMaps	LSeg (ViT-B-32)	0.448	0.304	0.198	0.330	0.485	0.260	0.175	0.359
	SEEM (Focal-L)	0.447	0.310	0.226	0.367	**0.503**	0.272	0.200	0.357
Ours (w/o secondary fusion)	SEEM (Focal-L)	0.434	0.306	0.215	0.344	0.448	0.261	0.179	0.337
Ours (w/ secondary fusion)	SEEM (Focal-L)	**0.507**	**0.360**	**0.238**	**0.394**	0.489	**0.284**	**0.202**	**0.380**

of matched pairs:

$$f_{\text{match}}(k) = \frac{|S_k|}{|M|}, \quad \text{AUC}^{\text{top}}_{\text{match}} = \int_0^1 f_{\text{match}}(x)\, dx. \tag{24}$$

Results: Fig. 4 presents qualitative results of our 2.5D open-vocabulary mapping, showing that small objects (e.g., bottles, laptops) are reliably captured rather than overlooked. Table 1 reports values at different top-k thresholds, the overall AUC^{top} score, and the average absolute difference between predicted and ground-truth instance counts. Since VLMaps is point-wise, masking its feature map with ground-truth instance masks produces a predicted instance count that matches the ground truth by construction. Therefore, only the matching-normalized metric is reported for VLMaps.

For the $\text{AUC}^{\text{top}}_{\text{match}}$, VLMaps performs markedly worse than both HOV-SG and our method, even with ground-truth masking. This is due to the 2D, point-wise storage of VLMaps features, where instance features within a grid cell are blended, particularly under vertical overlap. Between HOV-SG and our method, the scores are very similar, with HOV-SG slightly higher. Notably, HOV-SG uses 1024-dimensional features while our model uses 512-dimensional features, suggesting that our method achieves comparable performance with a more compact representation.

For the $\text{AUC}^{\text{top}}_{\text{gt}}$, our method attains higher scores, although the margin is modest, indicating comparable recall of ground-truth instances with a slight advantage for ours.

For the $\text{AUC}^{\text{top}}_{\text{pred}}$, our method exceeds HOV-SG by 18 percentage points. Because the denominator is the number of predicted instances, this gap highlights our robustness to spurious predicted instances caused by over-segmentation and redundant small-object predictions. HOV-SG suffers from low robustness due to SAM+CLIP's over-segmentation issues and poor integration of small-sized objects during the mapping process. In contrast, our method achieves robustness by avoiding over-segmentation through SEEM's panoptic segmentation model and maintaining better preservation of small-sized objects. Consistently, the per-scene mean absolute instance-count gap between predicted and ground-truth instances is 5.6 for our method versus 50.2 for HOV-SG.

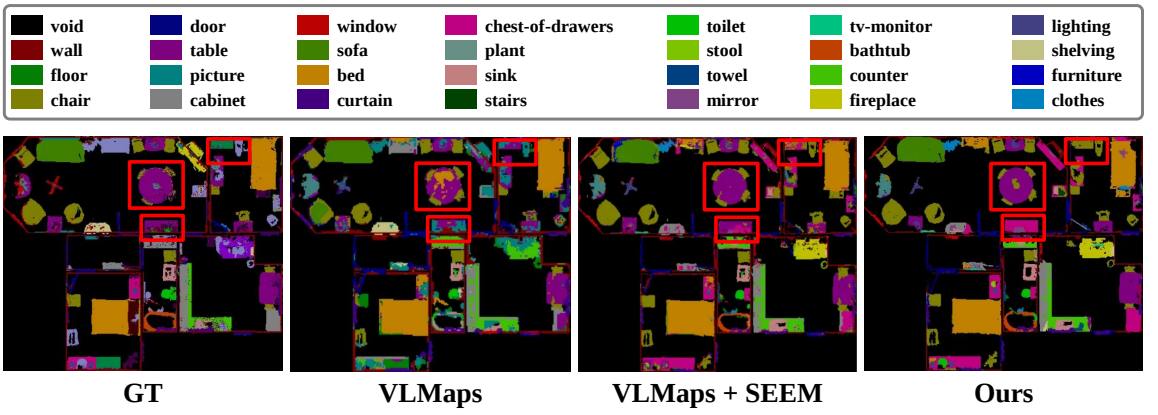
4.1.2. Additional Validation: Effectiveness of 2.5D Instance Mapping over 2D

In this section, we further investigate whether the 2.5D formulation provides advantages over a 2D mapping approach. To this end, we project our 2.5D map into a top-down 2D representation and compare it with purely 2D grid maps.

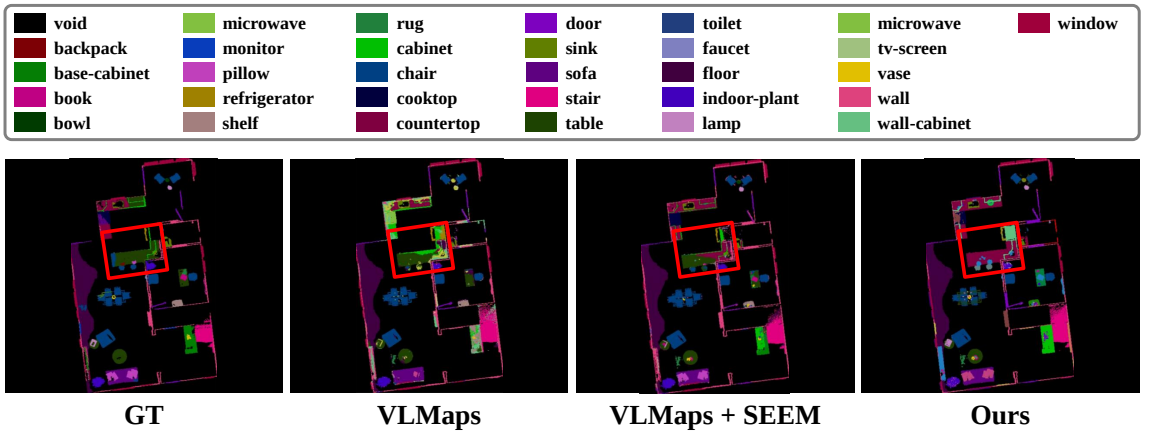
Baselines and Dataset: We compare against representative 2D projection baselines: VLMaps (LSeg, ViT-B-32), VLMaps+SEEM, which replaces LSeg with SEEM (Focal-L), and CLIP Map (CLIP, ViT-B-32), which encodes the environment using CLIP-derived representations. We also report our framework with and without secondary-instance fusion to verify the necessity of this step. Experiments are conducted on the MP3D and Replica datasets. For each dataset, four representative scenes are selected.

Metrics: Evaluation is conducted in a point-wise manner by comparing the predicted grid map with the ground-truth grid map. Specifically, we report four metrics. Pixel Accuracy (Acc) measures overall accuracy across all grid cells, Mean Accuracy (mAcc) is the mean class accuracy averaged over categories, Mean Intersection Over Union (mIoU) represents the mean IoU across categories, and Frequency Weighted Mean Intersection Over Union (F-mIoU) accounts for category frequency.

Results: The quantitative results are summarized in Table 2. Across most metrics, the performance follows the order: Ours, VLMaps+SEEM, VLMaps, and CLIP Map. Unlike models with pixel-level or object-level embeddings, CLIP Map uses an image-level embedding. As a result, the features stored in the map are intermixed across pixels and instances, which lowers performance. The comparison between VLMaps and VLMaps+SEEM highlights the



(a) 2D mapping results on scene 2t7WUuJeko7 of the MP3D.



(b) 2D mapping results on scene apartment_0 of the Replica.

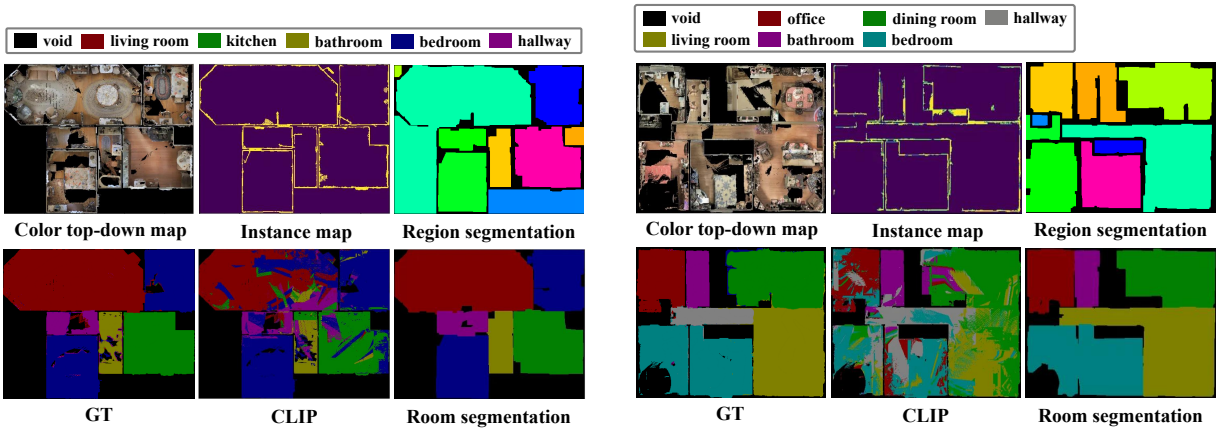
Figure 5: 2D mapping results of each method. The red boxes highlight that our method preserves objects as coherent instances without fragmentation. This demonstrates that the 2.5D mapping, when projected into a top-down view, mitigates the semantic ambiguity caused by vertically overlapping object features in conventional 2D projections.

Table 3
Evaluation of the room segmentation on the MP3D dataset.

Method	MP3D		
	Acc	mAcc	mIoU
CLIP	0.73	0.72	0.53
Ours	**0.89**	**0.86**	**0.75**

superior open-vocabulary capability of the SEEM visual encoder. Meanwhile, the improvement of our method over VLMaps+SEEM demonstrates the effectiveness of 2.5D mapping with instance-level integration. By preserving vertical distinctions that are collapsed in 2D projections, our method reduces semantic ambiguity and prevents erroneous category assignments. The qualitative results in Fig. 5 also confirm this trend, showing that our method produces more consistent object-level segmentations compared to the baselines.

In Fig. 5b, the region marked with a red box is labeled as "table" in the ground truth, while predicted as "countertop". This mismatch reflects the open-vocabulary nature of the task. The object is indeed a kitchen countertop, and such cases demonstrate that a single object may correspond to multiple semantically valid labels depending on the context.



(a) Room segmentation result on scene 2t7WUuJeko7 of the MP3D.

(b) Room segmentation result on scene RPmz2sHmrrY of the MP3D.

Figure 6: Room segmentation results on two scenes from the MP3D dataset. The top row shows the colorized top-down map, predicted instance map, and region segmentation. The bottom row compares the GT, CLIP-based semantic segmentation, and our final room-level segmentation output.

Overall, the comparison against 2D projection-based approaches confirms that our 2.5D framework not only improves accuracy but also provides more reliable and semantically coherent object-level representations in complex scenes.

4.2. Room Semantic Segmentation

The purpose is to assess whether room segmentation achieves spatially consistent partitions that align with structural cues, and whether the resulting regions retain accurate open-vocabulary semantic labels.

Baselines and Dataset: As a baseline, CLIP-based point features are projected onto the top-down map and used for pixel-wise room-type classification via similarity with predefined text prompts. Our proposed approach is quantitatively evaluated on the MP3D dataset, using two scenes.

Metrics: The evaluation is conducted using standard metrics including Acc, mAcc, and mIoU.

Results: Our method incorporates a structural room instance map, derived from architectural cues such as walls and doors, to guide region-wise label assignment based on CLIP. This structural constraint significantly reduces pixel-level noise and enhances spatial coherence in room segmentation. As summarized in Table 3, our method consistently outperforms the CLIP-based baseline across all evaluation metrics, with substantial improvements observed in mIoU. Fig. 6 illustrates qualitative segmentation results on two scenes from the MP3D dataset. We observe that CLIP-based segmentation suffers from semantic fragmentation and spatial inconsistency, often mislabeling adjacent areas due to the agent's limited viewpoint. In contrast, our method successfully corrects this issue by enforcing spatial contiguity and leveraging structural priors such as walls and doorframes, resulting in well-delineated and semantically coherent segmentation. These results confirm that our method effectively suppresses pixel-level noise and enforces spatial continuity by leveraging structural priors during label assignment.

4.3. Navigation

We evaluate navigation on the 30 most frequent categories in the dataset. For each scene and target category, one instance is uniformly sampled if multiple instances exist. To generate objective user queries, we prompt a language model with the ground-truth category, the ground-truth room label, and a representative RGB image of the target instance, where the object occupies the largest area. The resulting text is used as the query. See B for details. To assess robustness to query form, we define five types. (o) denotes category only. (o,r) denotes category and room. (o,d) denotes a category with a short description. (o,r,d) denotes category and room with a description. (a,r,d) denotes an abstract reference that omits the explicit category name with room and a description.

Table 4
Object retrieval success rates (O-SR) and navigation success rates (N-SR) from language queries on the HM3DSem dataset. For each scene and for every query type, we evaluate all categories that are present among the dataset's 30 frequent object categories. This yields approximately 15 trials per scene for each type. VLMaps is excluded from the object-retrieval comparison because its features are stored point-wise rather than instance-wise.

Method	Query type									
	(o)		(o,r)		(o,d)		(o,r,d)		(a,r,d)	
	O-SR [%]	N-SR [%]	O-SR [%]	N-SR [%]	O-SR [%]	N-SR [%]	O-SR [%]	N-SR [%]	O-SR [%]	N-SR [%]
VLMaps	–	8.09	–	5.09	–	2.51	–	3.84	–	1.25
HOV-SG	8.94	13.34	19.39	23.23	6.27	8.09	19.39	23.23	13.05	14.38
Ours	**14.93**	**17.51**	**21.19**	**25.51**	**20.43**	**22.53**	**21.27**	**25.60**	**20.01**	**21.34**

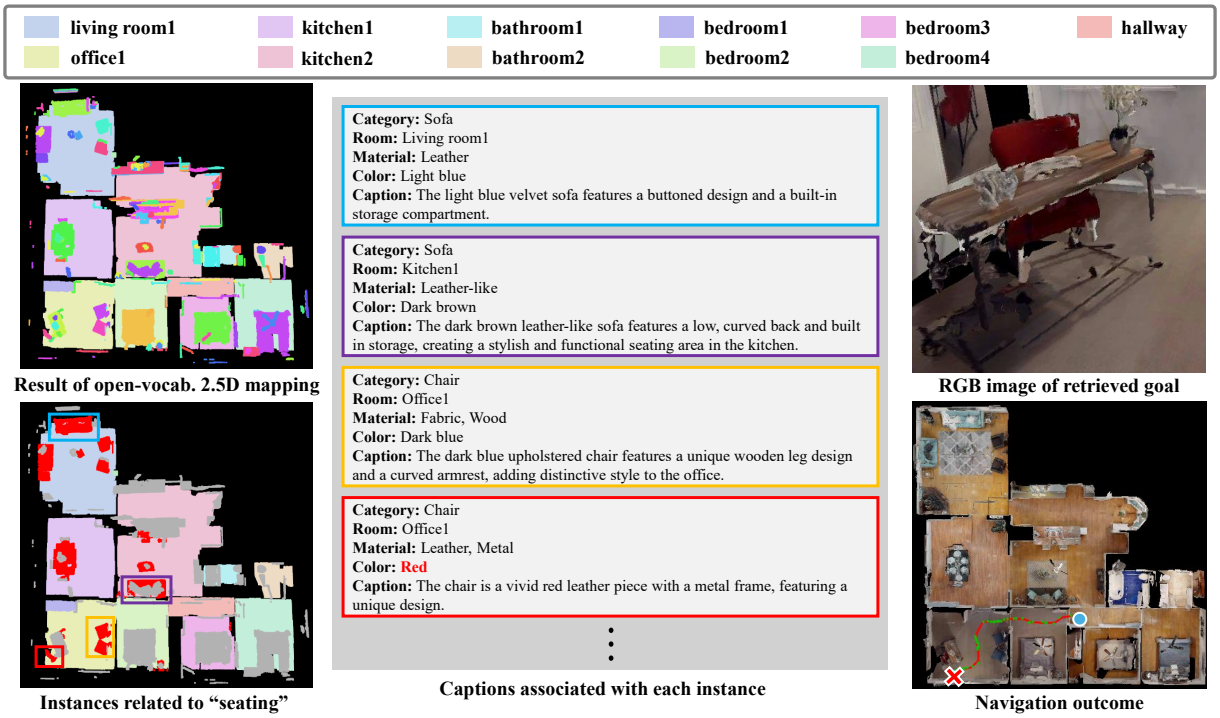


Figure 7: Navigation result on scene 00824 of the HM3DSem for the query, "Navigate to the red seating object in the office". The LLM selects the target instance by considering category, room-level information, and caption. Navigation is then executed toward the selected instance. In the visualization, the red cross marks the target location, and the blue circle indicates the starting point.

Baselines and Dataset: We compare VLMaps, HOV-SG, and our proposed method using the HM3DSem dataset. For multi-floor scenes, we evaluate only the first floor, as both VLMaps and our method assume a single-floor setting, and HOV-SG is restricted to the same floor for fair comparison.

Metrics: We measure two aspects. First, the object retrieval success rate asks whether the instance predicted from a query matches the intended ground-truth instance. A prediction is counted as correct if the intersection over union with the matched ground-truth instance exceeds 0.30, with the threshold chosen with reference to HOV-SG. For query types that do not uniquely specify the instance, we relax the criterion. For (o), we accept any ground-truth instance whose category matches the intended category. For (o,r), we accept any ground-truth instance whose category and room match the intended labels. In these two cases, we match the predicted mask to the ground-truth instance that yields the highest overlap and then check the labels. Second, the navigation success rate asks whether the agent reaches the target location. We declare success if the final stopping position lies within one meter of the ground-truth instance location.

Results: Fig. 7 illustrates the navigation result, and Table 4 reports the quantitative results. In Table 4, the best results are shown in bold. VLMaps stores features point wise rather than instance wise and does not produce instance proposals,

Table 5
Comparison of storage size. Results are measured in megabytes (MB) across five HM3DSem dataset scenes. For multi-floor scenes (00862, 00873, 00890), only the first floor is considered.

Dim.	Method	Avg.	Scene				
			00824	00829	00862	00873	00890
3D	HOV-SG	248	254	342	190	263	187
2D	VLMaps	118	186	99	179	68	57
2.5D	**Ours**	**10**	**11**	**10**	**13**	**9**	**9**

so computing object retrieval would require ground-truth masks or an external proposer. We therefore report navigation only for VLMaps.

On object retrieval, our method performs best on average across scenes and query types. HOV-SG extracts object and room keywords from the query, which leaves descriptive content unused, so the scores for (o) and (o,d) are nearly identical, and the same pattern holds for (o,r) and (o,r,d). In contrast, our method leverages instance-level captions. Adding descriptive phrases consistently improves retrieval, with the largest gains on (o,d) and (o,r,d). Comparisons against HOV-SG on (o,d) and (a,r,d) further indicate robustness to diverse query forms enabled by captioning. Whereas HOV-SG extracts category keywords from the query, our method lets the language model directly infer and rank candidates from predicted instance information, yielding stronger robustness to abstract references.

On navigation, our method also achieves the highest success rates on average across scenes and query types. Navigation success can exceed retrieval success when over segmentation or partial masks cause a prediction to miss the mask overlap threshold while still localizing close enough for the agent to stop within one meter. VLMaps exhibits lower navigation success than HOV-SG and our method due to the absence of instance wise representations and the lack of room and caption information required to handle diverse query forms. Consistent with object retrieval, the advantage of our method is most pronounced for (o,d) and (a,r,d). This indicates that storing descriptive information through captioning materially benefits language-conditioned navigation.

4.4. Storage Cost

Storage cost is an important consideration in mapping. We evaluated the storage requirements of VLMaps, HOV-SG, and our approach on five HM3DSem scenes, as reported in Table 5. Since both our method and VLMaps operate under a single-floor assumption, only the first floor was considered for multi-floor scenes (00862, 00873, 00890).

Despite being a 2D-based model, VLMaps required the largest storage due to its design of storing semantic features at every grid cell. In contrast, both HOV-SG and our method store features at the object level, significantly reducing memory usage. In particular, our method required even less storage than HOV-SG, benefiting from the structural compactness of its 2.5D representation. This shows that our method achieves competitive accuracy with superior storage efficiency.

5. Conclusion

This paper presents Instance-Enriched Semantic Maps, a unified framework for robust VLN with strengths in instance-level 2.5D rich-information mapping, robust query processing via LLM-based target selection, and memory-efficient 2.5D representation. Experimental validation demonstrates over 6% improvement in navigation success rates across various query types. Despite these advances, limitations remain. The use of multiple VLMs introduces computational overhead, restricting real-time deployment, and room segmentation relies on physical partitions, reducing effectiveness in open-plan environments. Therefore, future work will involve developing lightweight real-time processing approaches for dynamic environments and enhanced multi-frame fusion for small object detection, or exploring contextual room segmentation methods. Although real-world deployment is beyond the current experimental scope, the modular 2.5D representation and instance-level fusion framework can be readily adapted to real robotic systems that operate with estimated poses and RGB-D inputs.

A. Prompts for Caption Generation

As presented in Sec. 3.3, using the VLM-based methods, we used the following prompts to derive object-related descriptions. Our pipeline leverages two complementary models for instance captioning: LLaMA 3.2 Vision for

visual attribute extraction and GPT-4 for textual aggregation. While LLaMA 3.2 Vision provides rich object-level descriptions, it often introduces redundant wording and repeated expressions that hinder concise representation. To address this issue, we designed prompts instructing the model to eliminate unnecessary repetitions, and refine the language into a compact yet informative caption.

LLaMA 3.2 Vision was guided to produce descriptive captions for segmented instances, conditioned on representative keyframe images and the corresponding predicted category label. The employed prompt is shown in Fig. 8. The "user" content is shown in black and the "assistant" content in green. The resulting output is presented as depicted in Fig. 9.

GPT-4 is employed to refine instance-level captions generated by LLaMA 3.2 Vision. The model eliminates redundant expressions and extracts structured attributes for each instance. To guide this process, a tailored system prompt was used to explicitly instruct the model to summarize key visual traits into concise captions. The system prompt is shown in Fig.10.

As illustrated in Fig.3, this refinement step produces the final instance representations consisting of a room tag, a compact natural language description, and structured key–value attributes, which collectively support robust open-vocabulary navigation.

B. Prompts for Navigation

To evaluate navigation under diverse language conditions, we automatically generated queries using GPT-based prompting. Each query was conditioned on three inputs: a representative frame of the target instance, its instance-level category label, and the corresponding room category label. We defined five query types:

1) Object-based query: explicitly referring to the target object category
2) Room-based query: emphasizing the room context in which the object is located
3) Caption-based query: leveraging descriptive attributes extracted from instance-level captions
4) Mixed query: combining object, room, and caption information into a single instruction
5) Abstract query: indirect expressions of the object (e.g., "something to sit on") instead of explicit category names

Predefined lexical variants were employed to generate queries with various linguistic expressions. This design ensures that queries remain both diverse and systematically comparable across conditions. An example output is shown in Fig. 11, which illustrates the five types of queries.

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Q: There is one {class_name} in the scene. Describe its color, material, and one special feature. Don't repeat any word or phrase. Limit your response to 2 sentences maximum. Finish as soon as main traits are mentioned. Avoid repeating words, phrases, or patterns.

A: The rug is a dark grey, woven, and has a subtle pattern of rounded shapes. It has a soft, plush pile.

Figure 8: Example of the prompt used for instance-level captioning with LLaMA 3.2 Vision.

```
1: {
    "category": "cooktop",
    "room-type": "living room",
     "captions": [
       "The cooktop is black and made of stainless steel. It has a
        built-in instant-heat feature, allowing for quick and
        efficient cooking.",
       "The cooktop is black, made of cast iron, and features a
        built-in thermometer.",
      "The cooktop is black and made of stainless steel. It
       features a built-in induction cooking system."]
  }
```

Figure 9: Example of the result with LLaMA 3.2 Vision.

```
System: Identify and describe instances. Input and output must
be in JSON format. The input field 'captions' contains a list of
image captions aiming to identify the instance. Output 'color' as
a visual color of the identified instance. Output 'material'
making up the identified instance. Output 'caption' as a concise
description of the identified instance.
```

Figure 10: Example of the system prompt with GPT-4.

```
96: {
      "category": "toilet",
      "room-type": "bathroom",
      "queries": {
         "object": "Find a toilet characterized
                    by a white finish.",
         "room": "Look for a toilet located
                   in the bathroom.",
         "caption": "Find a toilet with a white color
                     and a rounded edge.",
         "mixed": "Search for a toilet characterized
                   by a white ceramic material and
                   a rounded edge located in the bathroom.",
         "mixed-abstract": "Look for something to sit on
                            featuring a white color and
                            a rounded edge located
                            in the bathroom."}
    }
```

Figure 11: Example of a navigation query.

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